

Model Estimates of Climate Change Impact on the Vegetation Cover and Atmospheric Carbon Sink

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Received January 27, 2004; in final form, August 3, 2004

Abstract—The vegetation cover of terrestrial ecosystems plays an important role in the regulation of energy and mass exchange in the atmosphere–land surface system. In particular, plants actively absorb atmospheric carbon dioxide, thereby reducing the greenhouse effect. An important quantitative characteristic of the vegetation cover is its net primary production. Changes in the terrestrial net primary production under possible climate changes in the 21st century are estimated. The computations are based on the dependence of net primary production on the surface radiation balance and evaporation. This dependence is derived using the relationship between the thermal conditions (annual total of effective daily mean surface temperature) and the radiation balance. Possible changes in the net primary production over Russia under a 1°C increase in the annual mean global surface temperature are estimated by using the atmosphere–ocean general circulation models ECHAM4/OPYC3 and HadCM3 and the intermediate-complexity IAP RAS climate model for anthropogenic scenarios of greenhouse gas changes. The results obtained are used to estimate the atmospheric carbon sink associated with changes in the vegetation cover.

INTRODUCTION

A distinctive feature of modern climatic warming is its unprecedented rate. In the 20th century, the annual mean global surface temperature increased by approximately 0.6°C [1]. During the period of recorded observations, the ten globally warmest years have fallen in the last two decades. Moreover, starting in the second half of the 1980s, anomalies of global annual mean surface temperature have been beyond the double range of rms deviations from the average regime in the first half of the 20th century [2]. According to paleoclimatic data, the rate of increase in the Northern Hemisphere temperature in the 20th century was much higher than the characteristic rates of its changes during the last two millennia [3].

An analysis of meteorological data has shown that the annual mean surface temperature over Russia increased by 0.9°C in the 20th century [4]. The surface temperature trends were found to be highly inhomogeneous in space. The strongest trend in annual mean surface temperature was observed in central Siberia, the Baikal regions, Transbaikalia, and the Amur and Maritime region. Warming was more noticeable in winter and spring: 4.7°C and 2.9°C per 100 years, respectively. The uncertainty in estimated tendencies for precipitation is much greater than that for surface temperature [1]. According to [4], the annual and seasonal (except for winter) precipitation amounts over Russia decreased, on the whole, in the second half of the 20th century. The most noticeable reduction in precipitation was observed in northeastern Russia, whereas a

weak tendency of precipitation growth was found over European Russia.

The inhomogeneity of climate changes predetermines a complex and ambiguous response of the vegetation cover to these changes. A tendency of climate warming can cause serious changes in the structure, species composition, and seasonal dynamics of terrestrial ecosystems. An increased global surface temperature, changed precipitation amounts, an increased duration of daily mean air temperatures no lower than 10°C, and other climatic characteristics may lead to degradation and fragmentation of the habitats of many plant species and to the occurrence of new existence conditions for individual vegetative communities and ecosystems [5].

In this paper, we analyze the response of vegetation cover to possible climate changes over the territory of Russia. Note that the estimation of the response of vegetation production processes to possible global climate changes is also motivated by the importance of physiological processes characteristic of the vegetation cover. Namely, these processes have a large effect on the net carbon flux in terrestrial ecosystems [6], promote the reduction in atmospheric carbon dioxide, and significantly affect the energy exchange intensity in the atmosphere–land surface system.

SIMULATION OF THE VEGETATION RESPONSE TO CLIMATE CHANGES

Modern studies of the response of vegetation in terrestrial ecosystems to climate changes are concerned

primarily with the analysis of changes in the spatial distribution of habitats of plant communities (*biomes*) and with the estimation of dynamics of vegetation zone boundaries. In particular, an approach is used in which the dynamics of a vegetative community is described in terms of the responses of constituting individuals (gap models) [7–9]. Models of this type involve a detailed description of biological features of plant growth, competitive relationships between plants, and the interrelation of the vegetation cover with climatic characteristics. These models are used mainly for analyzing the dynamics of forest biomes. A detailed description of physiological and ecological dependences in models of this kind requires that a large number of quantitative empirically determined characteristics of the processes under study be known, which complicates the use of these models for the global estimation of changes occurring simultaneously in several biomes.

Another widely used approach to the study of the vegetation response to climate changes is based on *bioclimatic schemes*, i.e., experimentally found interrelations between vegetation characteristics and some climatic parameters or their combination. The bioclimatic schemes most frequently used in models include Holdridge's scheme [10], Grigor'ev–Budyko's scheme [11], and Lieth's formula [12]. In Holdridge's scheme, different biomes are defined according to climatic variables, such as the mean annual biotemperature, the mean total annual precipitation, and the ratio of mean annual potential evapotranspiration to mean total annual precipitation. The Grigor'ev–Budyko scheme relates the geographic distribution of basic vegetation types to the surface radiation balance and the radiation dryness index. In Lieth's formula, the annual primary production is defined as a function of annual mean temperature and total annual precipitation. Potential global vegetation pattern under various scenarios of climate changes were assessed in studies based on bioclimatic schemes taking into account the relation of biome boundaries to climatic characteristics [13–15]. As a rule, it is assumed in such studies that vegetation zones are transformed instantly under a change in the climatic conditions; i.e., it is not taken into account that actual shifts of plant communities caused by global or regional climate changes proceed in processes (as long as tens or hundreds of years), which are known as successions [16].

The response of the global vegetation pattern to climate changes can also be assessed by using dynamic models. For example, a dynamic global vegetation model depending on climatic parameters was proposed in [17, 18]. For a given climatic scenario, this model produces a global vegetation pattern for all the biomes under consideration at any instant of time and estimates the times required for these biomes to reach an equilibrium state.

The above approaches to the study of the vegetation pattern response to climate changes are based on qual-

itative vegetation characteristics. However, a quantitative analysis is required for a deeper understanding of possible changes occurring in the intensity of diverse production and destruction processes in ecosystems under possible climate changes.

A quantitative characteristic of a plant community is its *net primary production* (NPP). This parameter shows the intensity of carbon sequestration in plants and is defined as the amount of organic matter accumulated over a certain time interval (usually one year) by a unit area of all aboveground and underground plants of a vegetative community.

Attempts to take into account the interaction of climate changes and vegetation production processes were made in a number of studies which analyzed possible changes in agroclimatic conditions, bioclimatic potential, and the crop yield of basic agricultural crops in various geographical regions for several climate change scenarios [19–23]. In [24, 25], a climate model [26] taking into account the carbon cycle was used to estimate possible regional changes in the net primary production over Russia, including the estimation of the connection of NPP variations with drought regimes.

Changes in the terrestrial net primary production can be assessed by using biospheric and ecosystem models. Examples of application of this class of models to the indicated purpose are given in [27, 28]. In [27], the CENTURY ecosystem model was used to estimate possible changes in the net primary production in boreal forests over central Canada in response to a doubled carbon dioxide level specified according to the general circulation climate model GISS. In [28], the net primary production in pine forests over southeastern France was assessed by using the biospheric model BIOME3 with four climate change scenarios specified according to the general circulation model ARPEGE. It should be noted that the application of biospheric and ecosystem models to the estimation of possible changes in the net primary production encounters a number of difficulties associated with the identification of some model characteristics for different plant communities.

In [29], a method was proposed to estimate the net primary production of plant communities from changes in the net surface radiation balance and evaporation. This method was used to compute possible changes in the net primary production over European Russia for three scenarios of quasi-stationary climate changes with a doubled carbon dioxide concentration.

Annual estimates of the net primary production obtained with several biospheric and ecosystem models were analyzed in [30]. It was noted that the results obtained with models involving a large number of physiological parameters and environmental characteristics (ecophysiological models) are comparable with the model results obtained by applying methods based on empirically determined dependences of net primary production on the annual means of climatic factors (bioclimatic models). The NPP values based on eco-

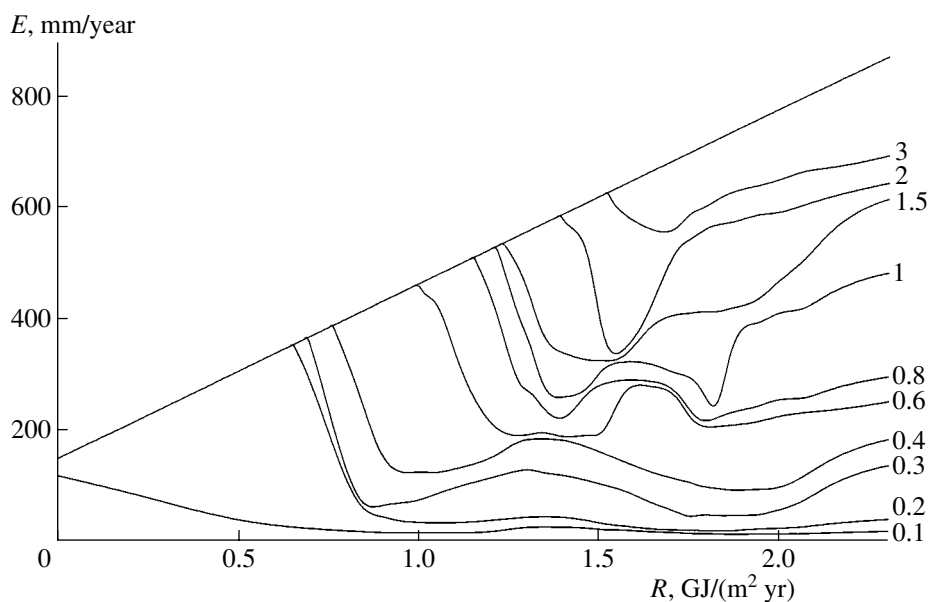


Fig. 1. Isolines of the net primary production ($\text{kg/m}^2 \text{ yr}$) over Russia displayed in the coordinates of evaporation (E , mm/yr) and net radiation balance (R , $\text{GJ/m}^2 \text{ yr}$).

physiological models exhibited a wider range of variations than those based on bioclimatic models. The analysis performed suggests that sufficiently simple models based on bioclimatic dependences can be used to reveal possible changes in the annual net primary production.

Obtained with the help of models with a different degree of detail in process description and with different spatiotemporal resolutions, the estimates of climate change impact on the net primary production in plant communities are fairly diverse. This testifies to a significant uncertainty in the existing estimates of possible vegetation changes over land surfaces. For this reason, a multilateral analysis of a possible response of terrestrial vegetation to climate changes requires a wide range of models, possible scenarios, and diagnostic methods.

METHOD FOR ESTIMATING THE RESPONSE OF NET PRIMARY PRODUCTION TO CLIMATE CHANGES

To estimate the net primary production response to possible climate changes, we used the output of the atmosphere–ocean general circulation climate models ECHAM4/OPYC3 (Max Planck Institute for Meteorology, Hamburg, Germany) [31, 32] and HadCM3 (UK Meteorological Office) [33, 34] over the period of 1860–2100. We also analyzed the output of the climate model (CM) developed at the Institute of Atmospheric Physics, Russian Academy of Sciences (IAP RAS) [2, 35, 36], which is an intermediate-complexity model [37]. In the numerical experiments, the atmospheric concentration of greenhouse gases was set equal to its observed values over the half period 1860–1990 and

followed the IS92a scenario over the half period 1991–2100 [38]. Note that, for the IAP RAS CM, we analyze the model results obtained under the IS92a scenario with a changed carbon dioxide level only. Regional changes in the net primary production over Russia were estimated under anthropogenic warming corresponding to a 1°C increase in the annual mean global surface temperature. It should be noted that the spatial change in climatic characteristics is rather nonuniform and depends significantly on a season in the case of the scenario under consideration. In particular, it should be expected that warming in the high latitudes over continents is maximal in winter, while the surface temperature changes in summer are less significant.

To determine the response of the net primary production in terrestrial ecosystems to climate changes, we used the model developed at the Laboratory of Mathematical Ecology of the IAP RAS [29]. This model is based on a bioclimatic scheme in which the net primary production of plant communities is a function of the net surface radiation balance R and evaporations E . To specify the bioclimatic scheme, these functional dependences were derived for the plain territory of Russia (mountain regions were not considered) from experimental data on the net primary production in plant communities of northern Eurasia [39] and from maps for radiation balance [40] and evaporation [41]. The resulting nomogram is presented in Fig. 1. By using this nomogram and piecewise cubic interpolation, the net primary production can be represented as a continuous function of two variables: net surface radiation balance and evaporation. Figure 2 displays the resulting dependence of NPP on R and E for regions of Russia. According to our computations, the deviations of the model's

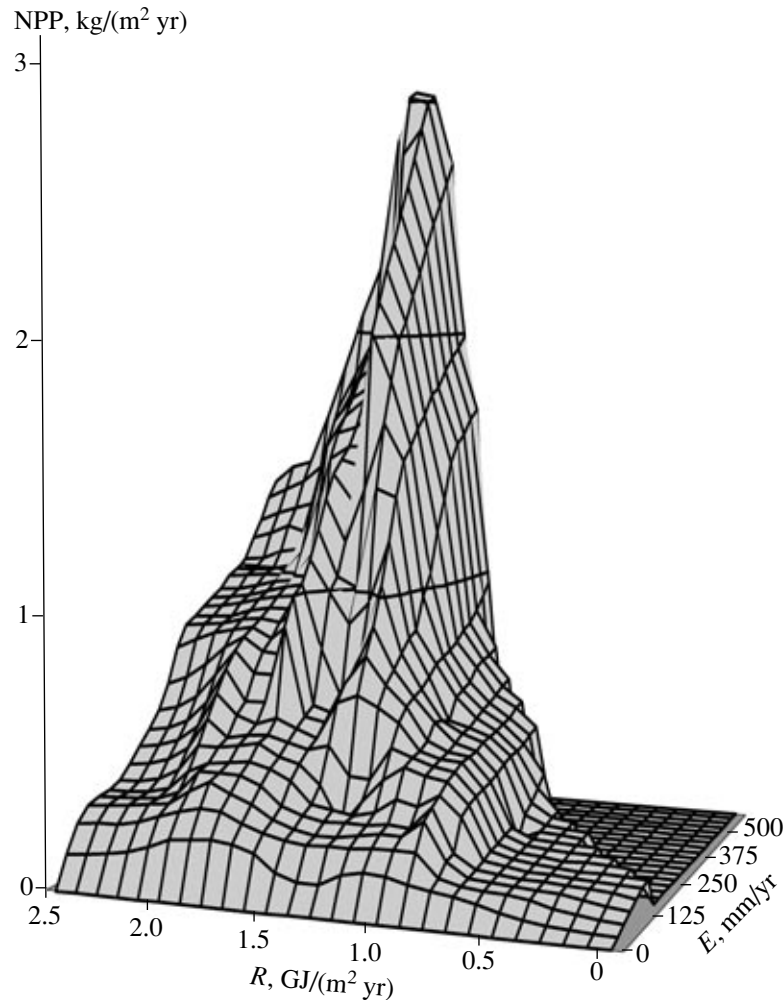


Fig. 2. Net primary production (NPP) as a function of the net surface radiation balance (R) and evaporation (E) for Russia.

NPP from its measured values are no more than 3% for tundra, 2% for coniferous forests, 10% for mixed forests, 16% for broadleaf forests, 35% for grasslands, 14% for steppes, and 9% for semideserts.

Thus, to calculate the response of the net primary production under a certain scenario of climate change, we have to estimate possible changes in evaporation and the net radiation balance at the surface. In this study, we used the evaporation change estimated from the output of climate models and the relation between the thermal conditions (the sum of effective temperatures, i.e., the annual sum of daily mean surface temperatures no lower than $10^{\circ}\text{C} - T_{\Sigma}$) and the net radiation balance R at the surface. Figure 3 presents the dependence between R and T_{Σ} over Russia as derived from the data in [40]. An analysis has shown that R increases insignificantly (from 0.7 to 0.8 GJ/m^2 per year) when T_{Σ} changes from 0 to 600°C . When T_{Σ} changes up to 2000°C , the values of R change from 0.8 to 1.5 GJ/m^2 per year, and a further increase in T_{Σ} up to 2800°C

reduces the growth of R . When T_{Σ} changes from 2800 to 3100°C , the values of R increase sharply from 1.8 to 2.1 GJ/m^2 per year. A further increase in T_{Σ} slightly affects the change in the net radiation balance: R remains virtually constant and is approximately equal to 2.0 GJ/m^2 per year. Thus, the interrelation between R and T_{Σ} for the territory of Russia is not linear. Since annual sums of effective daily mean surface temperatures can be estimated from outputs of climate models, the dependence between R and T_{Σ} can be used to estimate changes in the net radiation balance from changes in surface temperature.

ANALYSIS OF THE RESULTS AND CONCLUSIONS

Figure 4 shows possible changes in the net primary production under climate changes in the models ECHAM4/OPYC3, HadCM3, and IAP RAS CM caused by a 1°C increase in the annual mean global surface temperature according to anthropogenic scenar-

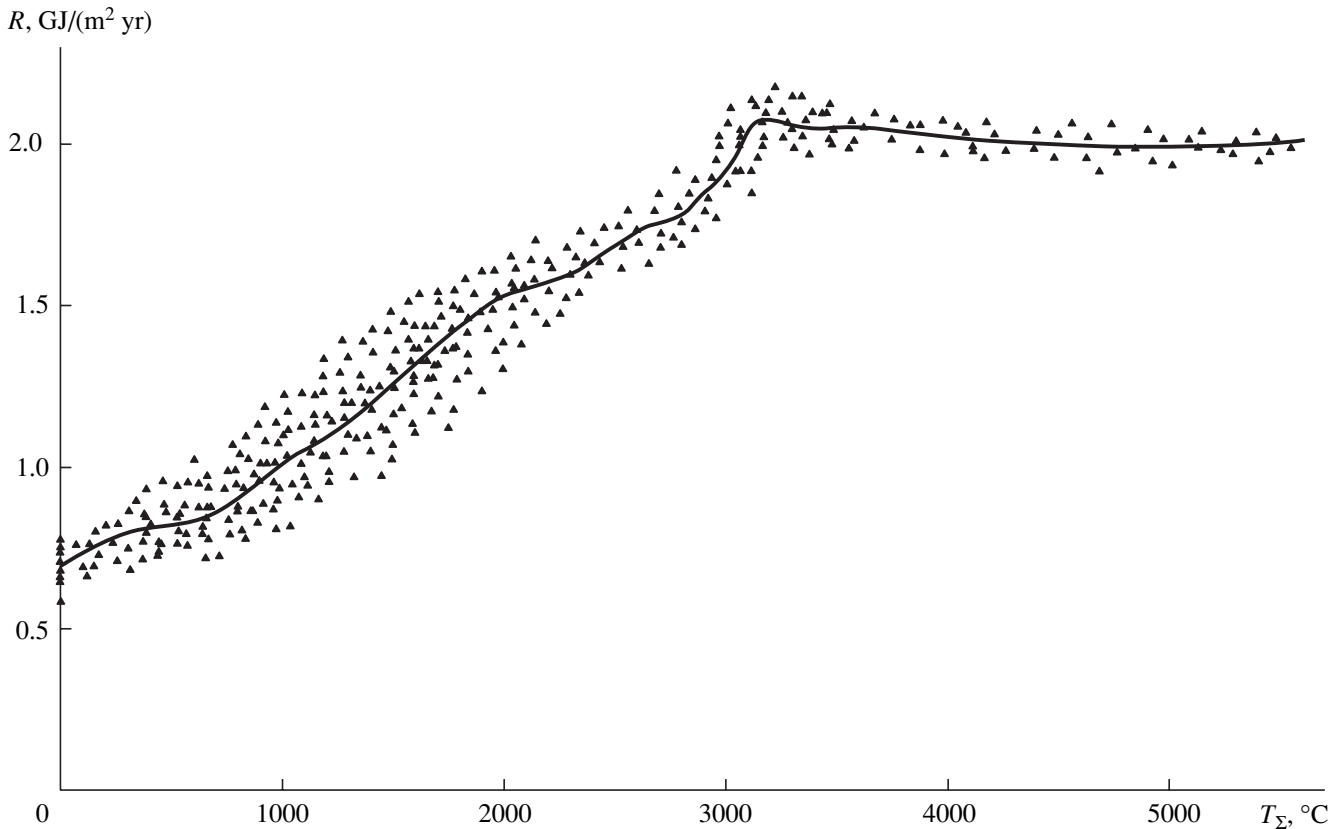


Fig. 3. Relation between the net surface radiation balance (R) and the annual sum of daily mean surface temperatures no lower than 10°C (T_{Σ}) for Russia (solid curve). The data of [40] are shown by triangles.

ios of global warming. The estimates obtained suggest that no large changes in the net primary production are expected over northeastern Russia and the northern regions of European Russia and Siberia. The change in NPP over these regions is found to range between -0.1 and 0.1 kg/m^2 per year. An increase in NPP by 0.1 – 0.4 kg/m^2 per year is possible in European Russia (southern Karelia and the Pechora–Onega Basin). An increase in NPP up to 0.4 kg/m^2 per year is also possible in several regions of western Siberia. According to the climatic scenarios in ECHAM4/OPYC3 and IAP RAS CM, an increase in the annual values of NPP by 0.1 – 0.4 kg/m^2 is obtained over extended middle Siberia (from the lower Vilyui and Lower Tunguska rivers in the north up to the upper Angara and Lena rivers in the south). All the climate change scenarios under consideration give a similar increase in NPP over Transbaikalia and the Far East.

A substantial increase in NPP (higher than 0.4 kg/m^2 per year) under the climatic scenarios of ECHAM4/OPYC3 and IAP RAS CM is possible over vast European Russia, western and middle Siberia, and the Far East. The HadCM3 scenario gives a similar increase in NPP over a much smaller territory (in Euro-

pean Russia, western Siberia, Transbaikalia, and the Far East).

A decrease in NPP by 0.1 – 0.8 kg/m^2 per year is obtained over a significant territory of European Russia (from the upper and middle Don River in the southwest to the lower Kama and Samara rivers in the east). According to the climatic scenarios of ECHAM4/OPYC3 and IAP RAS CM, a decrease in NPP by 0.3 – 0.4 kg/m^2 per year would be possible in the lower Volga region and the northern Caucasus. The ECHAM4/OPYC3 scenario also yields a possible decrease in NPP by 0.4 – 0.5 kg/m^2 per year over a southwest region (the lower Don River). According to the ECHAM4/OPYC3 and IAP RAS CM scenarios, an insignificant decrease in NPP (by up to 0.3 kg/m^2 per year) is obtained in southern Russia (from the Ural River Basin in the west to the Ob River Basin in the east). The HadCM3 scenario gives a reduction in NPP by 0.1 – 0.2 kg/m^2 per year over the lower Don region and by 0.1 – 0.5 kg/m^2 per year over a vast territory of western and middle Siberia (from the middle Irtysh and the lower Ob in the southwest to the middle and lower Lower Tunguska River in the northeast).

The results obtained suggest that the spatial distributions of NPP changes over Russia are fairly

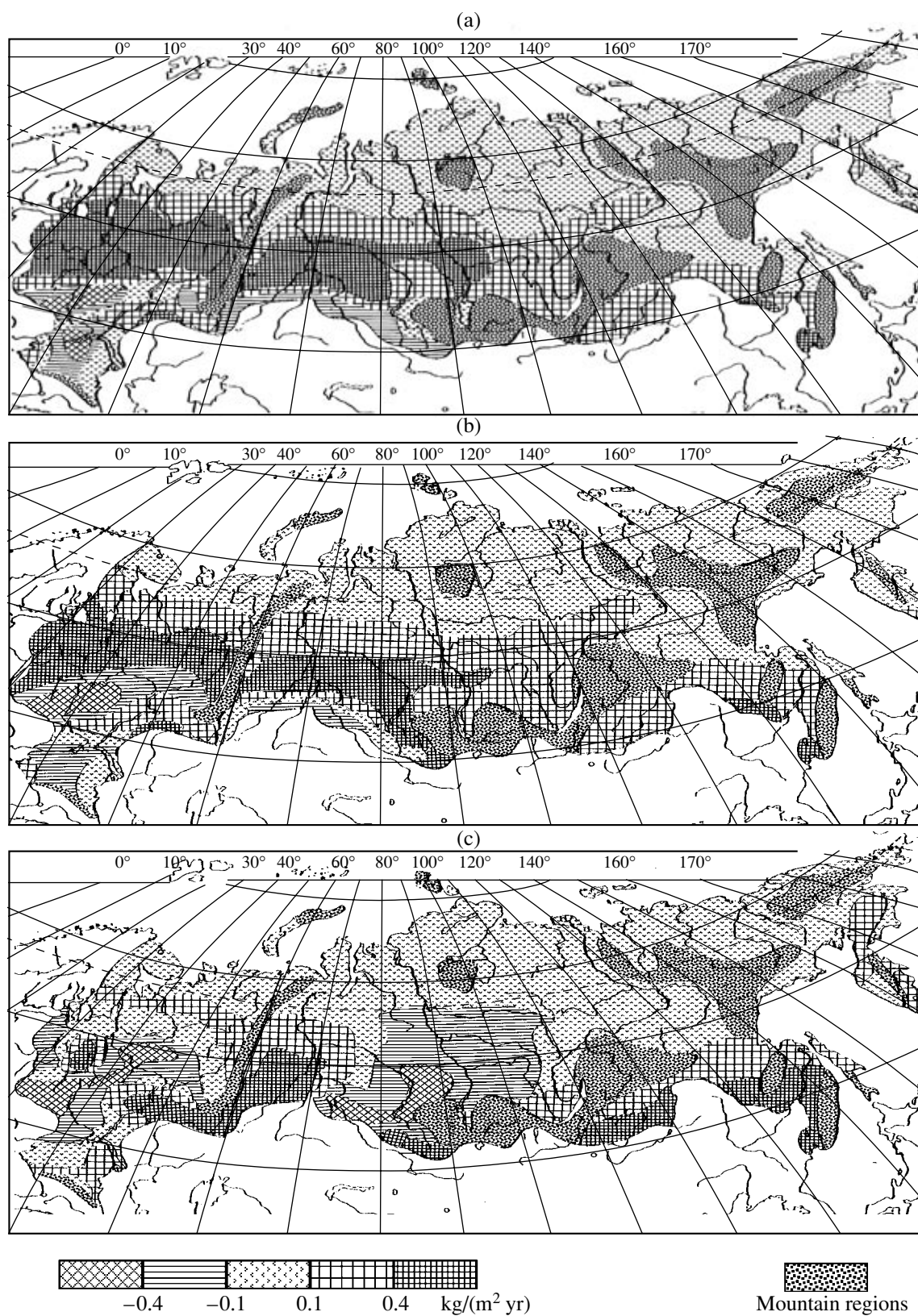


Fig. 4. Changes in the net primary production under climate changes associated with a 1°C increase in the annual mean global surface temperature in the (a) ECHAM4/OPYC3, (b) IAP RAS CM, and (c) HadCM3 models.

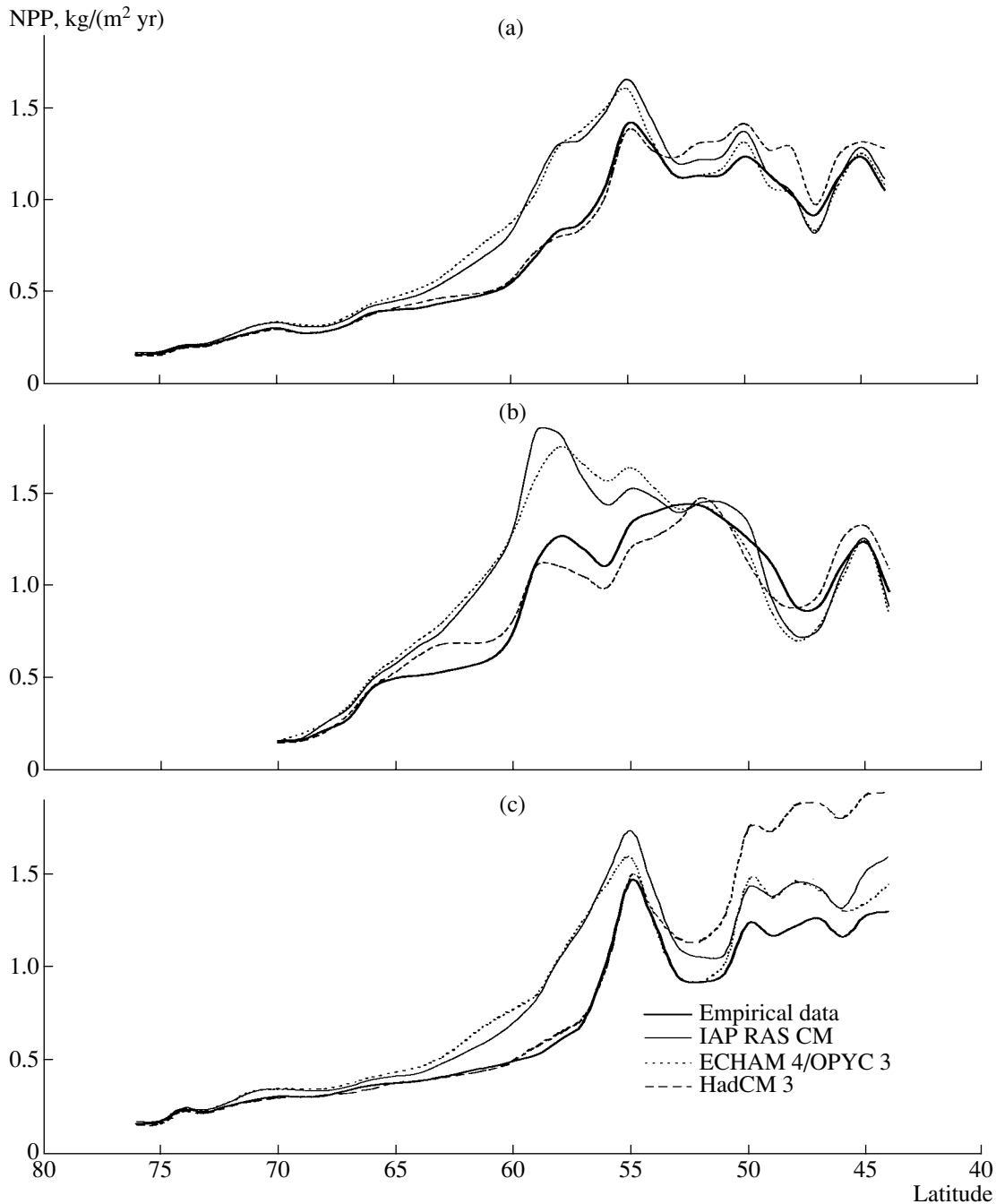


Fig. 5. Latitude distributions of NPP based on the empirical data over the second half of the 20th century (according to [39]) and under 1°C global warming according to the IAP RAS CM, ECHAM4/OPYC3, and HadCM3 climatic scenarios for (a) Russia, (b) European Russia, and (c) Siberia and the Far East.

close for the ECHAM4/OPYC3 and IAP RAS CM climatic scenarios, while the possible NPP changes under the HadCM3 scenario differ noticeably from those under the ECHAM4/OPYC3 and IAP RAS CM scenarios.

Figure 5 shows the latitude distributions of NPP derived from data over the second half of the 20th century for (a) all of Russia, (b) European Russia, and

(c) Siberia and the Far East. It can be seen in Fig. 5a that the empirical NPP maximum (about $1.4 \text{ kg}/\text{m}^2$ per year) over all of Russia is reached at about 55°N . The NPP values decrease sharply to $0.5 \text{ kg}/\text{m}^2$ between 55°N and 60°N . Farther to the north, the NPP decreases more gradually with increasing latitude. Between 55°N and 45°N , the latitude-averaged empirical NPP values over all of Russia are no less than $1 \text{ kg}/\text{m}^2$ per year.

A comparison of Figs. 5b and 5c shows that the empirical latitude dependences of NPP differ significantly for different regions of Russia. For example, the sharp NPP reduction north of 55° N observed in Fig. 5a is typical of Siberia and the Far East (Fig. 5c), while an additional NPP maximum north of 55° N appears over European Russia (Fig. 5b). The largest NPP values in European Russia are observed between 52°–53° N, whereas a pronounced local minimum of NPP occurs in these latitudes over Siberia and the Far East.

Along with the empirical data, Fig. 5 displays the estimates of possible NPP values over Russia obtained under a global warming of 1°C according to the climatic scenarios of ECHAM4/OPYC3, HadCM3, and IAP RAS CM. The changes in the model NPP values were estimated with respect to the empirical latitude distribution. For all of Russia (Fig. 5a), the largest NPP values in southern latitudes (up to roughly 53°–54° N) were obtained on the basis of the HadCM3 scenario. In this latitude range, the NPP values based on this scenario are larger than the empirical values for the second half of the 20th century. Farther to the north, NPP changes under the HadCM3 scenario are virtually absent. According to the ECHAM4/OPYC3 and IAP RAS CM climatic scenarios, the largest NPP changes are observed north of 55° N up to the subpolar latitudes. In the southern latitudes, the NPP changes over all of Russia according to the ECHAM4/OPYC3 and IAP RAS CM scenarios are less noticeable (especially for the ECHAM4/OPYC3 scenario) than those based on the HadCM3 model (Fig. 5a). In the 40th latitudes, the NPP values produced by the ECHAM4/OPYC3 and IAP RAS CM models (unlike HadCM3) are smaller than those inferred from the observations over the second half of the 20th century.

Substantial changes in NPP over European Russia were obtained north of approximately 53° N up to the subpolar latitudes (Fig. 5b). According to the ECHAM4/OPYC3 and IAP RAS CM scenarios, a significant increase in NPP under warming was obtained for this region. According to the HadCM3 scenario, the NPP values increase in the 60th latitudes and decrease south of them as compared with the empirical data for the second half of the 20th century.

For Siberia and the Far East (Fig. 5c), the NPP values increase in all latitude zones under the climatic scenario of IAP RAS CM. The ECHAM4/OPYC3 scenario gives similar estimates but with noticeably weaker NPP changes in a narrow midlatitude zone (52°–55° N). All the three climatic scenarios give a substantial increase in NPP south of 52° N in this region, with the largest NPP increase obtained under the HadCM3 scenario.

On the whole, the estimates based on the different model scenarios of 1°C global annual mean warming suggest a noticeable increase in NPP over a fairly extended latitude range in the territory of Russia. At the same time, we can see substantial regional differences

between the possible changes under the different model scenarios of climate changes. In particular, in the southern (40th) latitudes, the NPP values in Siberia and the Far East increase under the climatic scenarios of anthropogenic warming based on ECHAM4/OPYC3, HadCM3, and IAP RAS CM. Over European Russia, we observe oppositely signed NPP changes with a decreased NPP obtained in the latitude zone 47°–50° N according to all the climatic scenarios analyzed.

The vegetation cover of terrestrial ecosystems is an important part of the Earth's climate system. Climate changes lead to changes in the geographical and altitude boundaries of habitats of many plant species and to a redistribution of dominant species in ecosystems, which in turn change the net primary production in plant communities. Changes in vegetation characteristics of terrestrial ecosystems give rise to changes in land surface radiation characteristics and the atmosphere–land surface system as a whole. Energy exchange with the atmosphere and the hydrological and carbon regimes of the underlying regions also change [1, 42, 43]. From experimental data over the second half of the 20th century, the carbon sequestered in the vegetation cover over Russia is estimated to be about 4.4 Gt/year [44]. According to model estimates [30], the global net primary production in terrestrial ecosystems ranges from 46 to 62 Gt C/yr. The NPP value for terrestrial ecosystems in Russia is roughly 7–10% of its global value. It should be noted that the annual carbon sequestration in the Russian forest ecosystems, which occupy about 40% of the area of boreal and temperate forests in the Northern Hemisphere, is 2–2.5 Gt C/yr [44, 45]. This amounts to about 38% of the net primary production in forests of this type. The results obtained suggest that, under a global surface warming of 1°C, it can be expected that the annual carbon sequestration in plants over Russia would increase from roughly 7% (HadCM3 scenario) to 18% (ECHAM4/OPYC3 and IAP RAS CM scenarios) as compared with the observations over the second half of the 20th century.

ACKNOWLEDGMENTS

The authors are grateful to A.A. Karpenko for his assistance in processing the results of model computations of the climatic scenarios.

This work was supported by the Russian Foundation of Basic Research (project no. 02-05-64573) and the program of the Presidium of the Russian Academy of Sciences “Changes in the Natural Environment and Climate: Natural Disasters.”

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Translated by I. Ruzanova